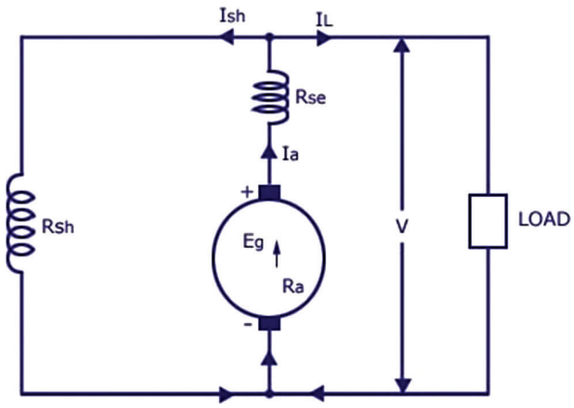


Fig 9.(a) *DC Long Shunt and*

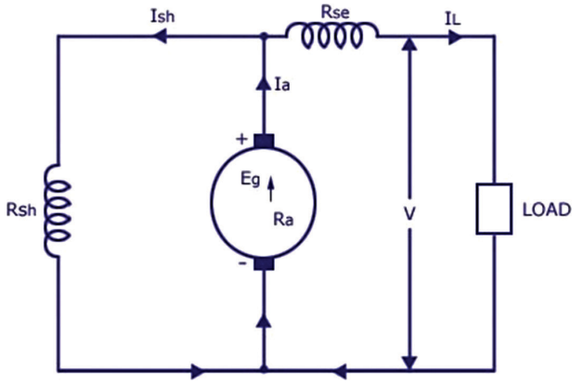


Fig 9.(b) *Short Shunt Compound Generator*

is connected in parallel with the series combination of series field winding and armature winding. (fig 7.9a)

In short shunt compound generator, the series field winding is connected in series with the parallel combination of armature and shunt field winding. (Fig 7.9 b)

Where,

I_a = Armature current
R_a =Armature resistance
I_L = Load current
I_{sh} =Shunt field current
I_{sc} =Series field current
R_{se} =Series field resistance
R_{sh} =shunt field resistance
V_L = load voltage
E_g = Generated voltage from armature.

7.5 EMF EQUATION OF DC GENERATORS

Let Ø = flux in webers.
Z = no. of armature conductors
N = speed in RPM
P = no of poles
A = no.of parallel paths
(A=P)- for lap winding
(A=2)- for wave winding

According to Faraday's law of electromagnetic induction, the value of generated

EMF, $E_g = \frac{d\text{Ø}}{dt}$

Flux cut/conductor in one revolution for 'P' poles 'dØ'=ØP

The time for one revolution 'dt'=60/N.

The value of induced emf per conductor,

$$E_g = \frac{d\text{Ø}}{dt} = \frac{\text{Ø}P}{\frac{60}{N}} = \frac{\text{Ø}PN}{60}$$

For 'Z' number of conductors, and for 'A' parallel paths,

$$E_g = \frac{\theta ZN}{60} X \frac{P}{A} \text{ volts}$$

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